

Frank Shan

✉ g3shan@uwaterloo.ca ☎ +1-263-881-6666 in frank-shan 🌐 knarfthekant 🌐 portfolio

HIGHLIGHTS

- Backend-leaning builder experienced in **distributed system design**, data modeling, and performance optimization, with a focus on correctness, scalability, and maintainability.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* Sep 2024 - Apr 2029
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* Sep 2025 - Apr 2026
Ziyutec Inc.

- Designed a modular **service-oriented architecture** (NestJS + PostgreSQL + Redis), enabling independent feature evolution, fault isolation, and scalable backend growth.
- Enforced **ACID-safe transactional workflows** for multi-step operations (e.g., onboarding, verification), improving data consistency and eliminating partial-write edge cases.
- Implemented **JWT-based authentication with refresh token rotation**, resolving session persistence issues and improving reliability of long-running user workflows.
- Maintained and extended a **multi-tenant ERP system** (60K+ LOC) supporting **1,000+ active projects**, improving stability and debugging efficiency in a legacy codebase.

Research Assistant | *Waterloo, ON* Feb 2026 - Present
University of Waterloo

- Designed experiments for a probabilistic programming system (PAL) compiling symbolic constraints into tensor operations, enabling **neuro-symbolic training** with strict structural guarantees.
- Developed reproducible experiment pipelines with controlled datasets and evaluation metrics, accelerating iteration on constraint-driven learning approaches.

PROJECTS

Mini VLLM | *Python, PyTorch* [github](#) 

- Built a lightweight **LLM inference engine** with continuous batching, request scheduling, and pluggable **paged KV-cache** architecture for efficient memory management.
- Implemented a custom **scheduler and block manager** to coordinate sequence execution and KV memory allocation, improving GPU utilization under concurrent workloads.

Goodline Inventory | *Python, Next.js, PostgreSQL* [github](#) 

- Built a **full-stack inventory system** with real-time stock tracking, order flows, and authentication, serving as a single source of truth for inventory operations.
- Designed relational schemas supporting per-unit tracking (serial numbers) and inbound/outbound linkage, ensuring traceability and data integrity.

Resume Builder | *Python, Jinja2, prompt_toolkit, LaTeX, LangGraph* [github](#) 

- Built a multi-step resume generation agent (LangGraph) that maps experience to job descriptions via structured reasoning and template-driven generation.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++, SQL, HTML, CSS

Frameworks: NestJS, Next.js, React, Express, FastAPI, LangGraph, PyTorch

Tools: AWS, PostgreSQL, Linux, Docker, Git

PROFESSIONAL CERTIFICATES

Stanford Online December 2025
XCS224N Natural Language Understanding [credential](#) 